OmniObject3D: Large-Vocabulary 3D Object Dataset for Realistic Perception, Reconstruction and Generation

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Figure 1. **OmniObject3D is a large-vocabulary 3D object dataset with massive high-quality real-scanned 3D objects and rich annotations.** It supports various research topics, *e.g.*, perception, novel view synthesis, neural surface reconstruction, and 3D generation.

Abstract

*Recent advances in modeling 3D objects mostly rely on synthetic datasets due to the lack of large-scale real-scanned 3D databases. To facilitate the development of 3D perception, reconstruction and generation in the real world, we propose **OmniObject3D**, a large-vocabulary 3D object dataset with massive high-quality real-scanned 3D objects. OmniObject3D has several appealing properties: **1) Large Vocabulary:** It comprises 6,000 scanned objects in 190 daily categories, sharing common classes with popular 2D datasets (*e.g.*, ImageNet and LVIS), benefiting the pursuit of generalizable 3D representations. **2) Rich Annotations:** Each 3D object is captured with both 2D and 3D sensors, providing textured meshes, point clouds, multi-view rendered images, and multiple real-captured videos. **3) Realistic Scans:** The professional scanners support high-quality object scans with precise shapes and realistic appearances. With the vast exploration space offered by OmniObject3D, we carefully set up four evaluation tracks: **a)** robust 3D perception, **b)** novel-view synthesis, **c)** neural surface reconstruction, and **d)** 3D object generation. Extensive studies are performed on these four benchmarks, revealing new observations, challenges, and opportunities for future research in realistic 3D vision.*

1. Introduction

Sensing, understanding, and synthesizing realistic 3D objects is a long-standing problem in computer vision, with rapid progress emerging in recent years. However, a majority of the technical approaches merely rely on unrealistic synthetic datasets [5, 16, 59] due to the absence of a large-scale real-world 3D object database. However, the appearance and distribution gaps between synthetic data and real data cannot be compensated for trivially, hence hindering their real-life applications. Therefore, it is imperative to equip the community with a large-scale and high-quality 3D object dataset from the real world, which can facilitate a variety of 3D vision tasks and downstream applications.

Recent advances partially fulfill the requirements while still being unsatisfactory. As shown in Table 1, CO3D [43] contains 19k videos capturing objects from 50 MS-COCO categories, while only 20% of the videos are annotated with accurate point clouds reconstructed by COLMAP [46]. Moreover, they do not provide textured meshes. GSO [13] has 1k scanned objects while covering only 17 household classes. AKB-48 [26] focuses on robotics manipulation with 2k articulated object scans in 48 categories, but the focus on articulation leads to a relatively narrow semantic distribution, failing to support general 3D object research.

To boost the research on general 3D object understanding and modeling, we present **OmniObject3D**: a large-vocabulary 3D object dataset with massive high-quality real-scanned 3D objects. Our dataset has several appealing properties: **1) Large Vocabulary:** It contains 6,000 high-quality textured meshes professionally scanned from real-world objects, which, to the best of our knowledge, is the largest among the real-world 3D object datasets. It comprises 190 daily categories, which shares common classes with popular 2D and 3D datasets (*e.g.*, ImageNet [12], LVIS [20], and ShapeNet [5]), incorporating most daily object realms (See Fig. 1 and Fig. 6). **2) Rich Annotations:** Each 3D object is captured with both 2D and 3D sensors, providing textured 3D meshes, sampled point clouds, posed multi-view images rendered by Blender [10], and real-captured video frames with foreground masks and COLMAP camera poses. **3) Realistic Scans:** The object scans are of high fidelity thanks to the professional scanners, bearing precise shapes with geometric details and realistic appearance with high-frequency textures.

Taking advantage of the vast exploration space offered by OmniObject3D, we carefully set up four evaluation tracks: **a)** robust 3D perception, **b)** novel-view synthesis, **c)** neural surface reconstruction, and **d)** 3D object generation. Extensive studies are performed on these benchmarks: First, the high-quality real-world point clouds in OmniObject3D allow us to perform *robust 3D perception* analysis on both out-of-distribution (OOD) styles and corruptions, the two major challenges in point cloud OOD generalization, at the same time. What's more, we provide massive 3D models with precise 3D ground truth and multi-view images for *novel-view synthesis* and *neural surface reconstruction*. The broad diversity in shapes and textures offers a comprehensive training and evaluation source for both scene-specific and generalizable algorithms. Finally, we equip the community with a database for in-the-wild *3D object generation*, which pushes the boundary of existing state-of-the-art generation methods to real-world 3D objects. The four benchmarks reveal new observations, challenges, and opportunities for future research in realistic 3D vision.

Table 1. A comparison between OmniObject3D and other commonly-used 3D object datasets. R^{lvis} denotes the ratio of the 1.2k LVIS [20] categories being covered.

Dataset	Real	Full 3D	Video	# Objs	# Cats	R^{lvis} (%)
ShapeNet [5]		✓		51k	55	4.1
ModelNet [59]		✓		12k	40	2.4
3D-Future [16]		✓		16k	34	1.3
ABO [9]		✓		8k	63	3.5
Toys4K [49]		✓		4k	105	7.7
CO3D [43]	✓		✓	19k	50	4.2
DTU [1]	✓	✓		124	NA	0
ScanObjectNN [52]	✓			15k	15	1.3
GSO [13]	✓	✓		1k	17	0.9
AKB-48 [26]	✓	✓		2k	48	1.8
Ours	✓	✓	✓	6k	180	10.8

2. Related Works

3D Object Datasets. The acquisition of a large-scale realistic 3D database is usually an expensive and challenging task. Many widely-used 3D datasets prefer to collect synthetic CAD models from online repositories: ShapeNet [5] collects 51,300 3D CAD models into 55 categories within the WordNet [35] taxonomy. ModelNet40 [59] consists of 12,311 CAD-generated meshes in 40 categories. Recent works [9, 16] introduce high-quality CAD models with rich geometric details and informative textures, where 3D-FUTURE [16] collects 16,563 objects in 34 furniture categories and ABO [9] consists of 7,953 products across 63 household categories. Toys4K is comprised of 4,179 objects in 105 categories which supports the exploration of generalizable 3D networks. Due to the inevitable gaps between synthetic and real objects, the community is always eager for a large-scale 3D object dataset in the real-world domain. DTU [1] and BlendedMVS [63] are photo-realistic datasets designed for multi-view stereo benchmarks, which contain only 124 and 113 scenes without category annotations, respectively. ScannObjectNN [52] is a real-world point cloud object dataset based on scanned indoor scenes, which has around 15,000 objects with colored point clouds categorized into 15 categories. However, the point clouds are incomplete and noisy, with multiple objects co-existing in one scene. GSO [13] has 1,030 scanned objects with fine-grained geometries and textures in 17 categories of household items, and AKB-48 [26] focuses on robotics manipulation with 2,037 articulated object models in 48 articulated object categories. However, the relatively narrow semantic scope of both GSO and AKB-48 hinders their applications

CVPR
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Figure 2. **Semantic distribution of the OmniObject3D dataset.** It covers 190 daily categories with a long-tailed distribution, sharing common classes with popular 2D and 3D datasets.

for more general 3D research. CO3D [43] contains 19,000 object-centric videos, while only 20% of them are annotated with accurate point clouds reconstructed by COLMAP [46], and they do not provide meshes or textures. In contrast, the proposed OmniObject3D dataset comprises 6,000 3D objects scanned by professional devices with meshes, textures, and multi-view photos in 190 categories, fulfilling the requirements of a wide range of research objectives. A detailed comparison between OmniObject3D and other commonly-used datasets is presented in Table 1.

Neural Radiance Field. Neural radiance field (NeRF) [34] represents a scene with a fully-connected deep network (MLPs), which takes in hundreds of sampled points along each camera ray and outputs the predicted color and density. Novel views of the scene are synthesized by projecting the colors and densities into an image via volume rendering. Inspired by the success of NeRF, a massive follow-up effort has been made to improve its quality [3, 4, 33, 54], and efficiency [6, 15, 36, 50]. A branch of works [7, 29, 43, 56, 66] has also explored the generalization ability of NeRF-based frameworks. PixelNeRF [66], MVSNeRF [7], IBRNet [56], and NeuRay [29] reconstruct the radiance field with a mere forward pass during inference via training on cross-scenes. NeRFormer [43], IBRNet [56], and GNT [53] leverage Transformers for generalizable NeRF.

Neural Surface Reconstruction. Implicit Neural Representations (INR) [2, 8, 22, 30, 32, 39, 45, 48, 51, 68] of 3D object geometry and appearance with neural networks have attracted increasing attention in recent years. Some approaches [23, 27, 37, 65] regard the color of an intersection point between the ray and the surface as the rendered color, namely surface rendering, and they typically rely on accurate object masks. Another trend of recent approaches [11,38,55,58,64] proposes to leverage neural radiance field with implicit surface representations like Signed Distance Function (SDF) for higher-quality and mask-free surface reconstruction from multi-view images. NeuS [55], VolSDF [64] reconstruct implicit surfaces with SDF-based volume rendering scheme, and Voxurf [58] leverages an explicit volumetric representation for acceleration. Since dense camera views of scenes are sometimes not available, SparseNeuS [31] and MonoSDF [67] explore surface reconstruction from sparse views. The former exploits generalizable priors cross scenes for a generic surface prediction, while the latter takes advantage of the estimated geometry cues predicted by pretrained networks.

OmniObject3D can serve as a large-scale benchmark with realistic photos and meshes for both training and evaluation. It bears a large vocabulary and high diversity in shape and appearance, offering an opportunity for pursuing more generalizable and robust novel view synthesis and surface reconstruction methods.

Related works for **Robust Point Cloud Perception** and **3D Generation** can be found in supplementary materials.

3. The OmniObject3D Dataset

In this section, we describe the data collection, processing, and annotation pipeline of OmniObject3D. We will also introduce its properties and statistics, and conduct a comprehensive comparison with other widely-used 3D datasets.

3.1. Data Collection, Processing, and Annotation

Category List Definition. In order to collect a large amount of 3D objects that are both commonly-distributed and highly diverse, we first pre-define a category list according to several popular 2D and 3D datasets [5, 12, 20, 24, 25, 47, 59]. We cover most of the categories that lie within the application scope of the scanners and also dynamically expand the list with new classes out of the current list during the collection process. We end up with 190 widely-spread categories, which ensures a library with rich texture, geometry, and semantic information.

Object Collection Pipeline. We then collect a variety of objects from each category and use professional 3D scanners to obtain high-resolution textured meshes. Specifically, we use the Shining 3D scanner [1] and Artec Eva 3D scanner [2] for objects in different scales, respectively. The scanning time varies with the properties of the object: it takes around 15 minutes to scan a small rigid object with a simple geometry (*e.g.*, an apple, a toy, *etc.*), while it takes up to an hour to obtain a qualified 3D scan for the non-rigid, the complex, or large objects (*e.g.*, a bed, a kite, *etc.*). For around 10% of the objects, we conduct common manipulations (*e.g.*, taking a bite, cutting in pieces) to conform the natural instincts of them. The 3D scans can faithfully retain the real-world scale of each object, but their poses are not strictly aligned. We thus pre-define a canonical pose for each category and manually align the objects within a category. We then check the quality of each scan, and around 83% high-quality ones out of the total collection are finally reserved in the dataset.

Image Rendering and Point Cloud Sampling. To support a variety of research topics like point cloud analysis, neural radiance fields, and 3D generation, we render multi-view images and sample point clouds based on the collected 3D models. We use Blender [10] to render object-centric and photo-realistic multi-view images, together with accurate camera poses. The images are rendered from 100 random viewpoints sampled on the upper hemisphere at 800×800 pixels. We also produce high-resolution mid-level cues like depth and normal for more research use. We then uniformly sample multi-resolution point clouds from each 3D model using the Open3D toolbox [69], with $2^n, n \in 10, 11, 12, 13, 14$ points in each point cloud, respectively. Besides the multi-modality data existing in the dataset, we also provide the users with a data generation pipeline. One can easily obtain new data with self-defined camera distributions, lightning, and point sampling methods to meet different research requirements.

Video Capturing and Annotation. After scanning each object, we capture a video of it. The object is placed on or beside a calibration board, and each video covers a full 360° range around it. Square corners on the calibration board can be recognized by the QR Codes beside it, and we then filter out the blurry frames whose recognized corners are less than 8. We uniformly sample $N = \min(200, N^c)$ frames from the N^c frames left, and then the COLMAP [32], a well-known SfM pipeline, is applied to annotate the frames with camera poses. Finally, we use the scales of the calibration board in both the SfM coordinate space and the real world to recover the absolute scale of the SfM coordinate system. We also develop a two-stage matting pipeline based on the U^2Net [42] and FBA [14] matting model to generate the foreground masks for all the frames. Please refer to supplementary materials for more implementation details.

[1] https://www.einscan.com/
[2] https://www.artec3d.cn/

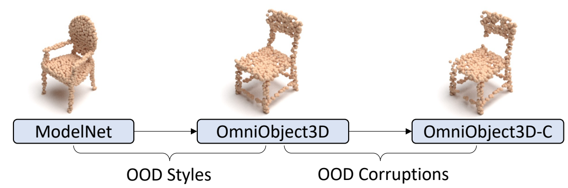

Figure 3. OmniObject3D provides the first clean real-world point cloud object dataset, and allows fine-grained analysis on robustness to OOD styles and OOD corruptions."-C": corrupted by common corruptions described in [44]

3.2. Statistics and Distribution

With 6,000 3D models in 190 categories, OmniObject3D exhibits a long-tailed distribution with an average of around 30 objects in each category, as shown in Fig. 2. It shares many common categories with several famous 2D and 3D datasets [5, 12, 20, 24, 25, 47, 59], *e.g.*, it covers 130 categories in LVIS [20] and 85 in ImageNet [12]. Most of the categories are covered by the OpenImage [24] image-level labels. It also bears a huge diversity in shapes and appearances. The vast semantic and geometrical exploration spaces enable a wide range of research objectives.

4. Experiments

4.1. Robust 3D Perception

In this section, we show how OmniObject3D can contribute to the robustness analysis of point cloud perception. To generalize point cloud perception models in real-world applications, two critical out-of-distributuion (OOD) challenges have to be addressed: 1) OOD styles, *e.g.*, different styles between synthetic CAD models and real-world objects; 2) OOD corruptions, *e.g.*, random point jittering or missing due to sensory inaccuracy.

Existing robustness evaluation utilizes clean synthetic datasets, *e.g.*, ModelNet [59], for training and sets up two kinds of test sets for evaluation:

1) Noisy real-world datasets, *e.g.*, ScanObjectNN [52], which are cropped from real-world scenes. They are employed to measure the robustness of the sim-to-real domain gap. However, the gap couples both OOD styles and OOD corruptions at the same time as the cropped point clouds are always noisy, making it impossible to analyze the two robustness challenges independently.

2) Corrupted synthetic datasets, *e.g.*, ModelNet-C [44], which are artificially perturbed on top of clean synthetic datasets. The evaluation allows for detailed corruption analysis, but they do not reflect the robustness to OOD styles.

None of the existing robustness benchmarks allows analyzing the robustness to both OOD styles and OOD corruptions in fine granularity. OmniObject3D, on the other hand, as the first clean real-world point cloud object dataset, can help to address the issue. For models trained on Mod-

Table 2. Point cloud perception robustness analysis on OmniObject3D with different architecture designs. Models are trained on the ModelNet-40 dataset, with OA_{Clean} to be their overall accuracy on the standard ModelNet-40 testset. OA_{Style} on OmniObject3D evaluates the robustness to OOD styles. mCE on the corrupted OmniObject3D-C evaluates the robustness to OOD corruptions. Full results are presented in the supplementary materials.

	OA_{Clean} ↑	OA_{Style} ↑	mCE ↓
DGCNN [57]	0.926	0.448	1.000
PointNet [40]	0.907	0.466	0.969
PointNet++ [41]	0.930	0.407	1.066
RSCNN [28]	0.923	0.393	1.076
SimpleView [18]	**0.939**	0.476	0.990
GDANet [62]	0.934	0.497	**0.920**
PAConv [61]	0.936	0.403	1.073
CurveNet [60]	0.938	**0.500**	0.929
PCT [19]	0.930	0.459	0.940
RPC [44]	0.930	0.472	0.936

elNet, we first evaluate their performance on OmniObject3D to examine OOD-style robustness. Then, we create OmniObject3D-C by corrupting OmniObject3D with common corruptions described in [44] to examine the OOD-corruption robustness. We show a complete robustness evaluation scheme in Fig. 3. For evaluation metrics, we use the overall accuracy (OA) on OmniObject3D to measure the OOD-style robustness and use DGCNN normalized mCE [44] to measure the OOD-corruption robustness.

We benchmark 10 state-of-the-art point cloud perception models in Table 2. We observe that 1) performance on clean test set has little correlation to OOD-style robustness. For example, SimpleView [18] achieves the best OA_{Clean} but mediocre OA_{Style}; 2) Advanced point grouping, *e.g.*, curve-based point grouping in CurveNet [60] and frequency-based point grouping in GDANet [62], are robust not only to OOD corruptions as pointed out in [44], but also to OOD styles; 3) OOD style + OOD corruption is a more challenging setting. In particular, RPC, the most robust architecture to OOD corruptions [44], shows inferior mCE. In summary, robust point cloud perception models against both OOD styles and OOD corruptions are still under-explored. The OmniObject3D dataset sheds new light on a comprehensive understanding of point cloud perception robustness.

4.2. Novel View Synthesis

In this section, we study several representative methods on OmniObject3D for novel view synthesis (NVS) in two settings: 1) training on a single scene with densely captured images, which is the standard setting for NeRF [34]; 2) learning priors across scenes from our dataset to explore the generalization ability of NeRF-style models.

Single-Scene NVS. Under this setting, we select three objects in each category for the experiments. We involve the original NeRF [34], a famous extension of it called mip-NeRF [3], and a voxel-based system named Plenoxels [15] for comparison. The results are shown in Table 3. We find that Plenoxels achieve the best performance on average for all the three metrics. There exists a clear margin for LPIPS between Plenoxels and the other two methods since voxel-based methods are especially good at modeling high-frequency appearance. We also present the standard deviation (SD) of results across all the training samples for each method, where Plenoxels are relatively unstable compared to NeRF and mip-NeRF. We observe that this voxel-based method introduces artifacts when encountering concave geometry (*e.g.*, bowls, chairs), and it suffers from an inaccurate density field modeling when the foreground object is dark. MLP-based methods are relatively robust against these difficult cases. In a nutshell, our dataset provides a library with a variety of shapes and appearances, allowing a comprehensive evaluation for different NVS methods in different cases. See the supplementary for more detailed analysis and visualization.

Table 3. Single-scene novel view synthesis results and their standard deviation (SD) across the training set.

Method	PSNR (↑) / SD	SSIM (↑) / SD	LPIPS (↓) / SD
NeRF [34]	34.01 / 3.46	0.953 / 0.029	0.068 / 0.061
mip-NeRF [3]	39.86 / 4.58	0.974 / 0.013	0.084 / 0.048
Plenoxels [15]	**41.04** / 6.84	**0.982** / 0.031	**0.030** / 0.031

Table 4. Cross-scene novel view synthesis results on 10 categories. 'Cat.' and 'All*' denote training on each category and training on all categories except the 10 test ones, respectively.

Method	Train	PSNR (↑)	SSIM (↑)	LPIPS (↓)	$\mathcal{L}_1^{\text{depth}}$ (↓)
MVSNeRF [7]	All*	17.49	0.544	0.442	0.193
	Cat.	17.54	0.542	0.448	0.230
	All*-ft.	**25.70**	**0.754**	**0.251**	0.081
	Cat.-ft.	25.52	0.750	0.264	**0.076**
pixelNeRF [66]	All*	22.16	0.692	0.342	0.109
	Cat.	20.65	0.676	0.348	0.195

Cross-Scene NVS. We conduct extensive experiments on novel view synthesis from sparse inputs by pixelNeRF [66] and MVSNeRF [7] as shown in Table 4. We select 10 categories with the most various scenes as the test set. All metrics are averaged over 300 images. In the generalization setting, although not trained on the test category, $\text{MVSNeRF}_{\text{All*}}$ is comparable to $\text{MVSNeRF}_{\text{Cat.}}$ and $\text{pixelNeRF}_{\text{All*}}$ even outperforms $\text{pixelNeRF}_{\text{Cat}}$ in all terms of visual metrics, especially on regular-shaped objects such as squash and apple. It confirms that OmniObject3D serves as an information-rich dataset that is beneficial for obtaining a strong generalizable prior on unseen scenes. Moreover, it is worth noting that each method w/ 'All*' generates better underlying depth than that w/ 'Cat.', inferring generalizable methods can implicitly learn geometric cue though only trained from appearance in our dataset. It is reasonable that

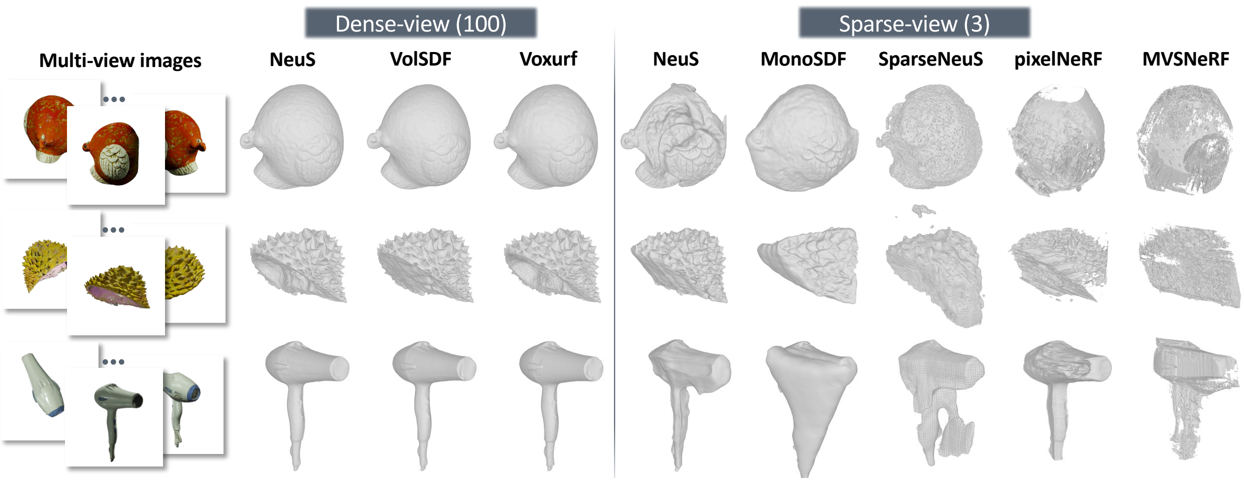

Figure 4. **Neural surface reconstruction results for both dense-view and sparse-view settings.**

Table 5. Single-scene dense-view surface reconstruction results.

Method	Chamfer Distance $\times 10^3$ ($\downarrow$)			
	Hard	Medium	Easy	Avg
NeuS [55]	9.26	5.63	3.46	6.09
VolSDF [64]	10.06	4.94	2.86	5.92
Voxurf [58]	9.01	4.98	2.58	**5.49**
Avg	9.44	5.19	**2.97**	5.83

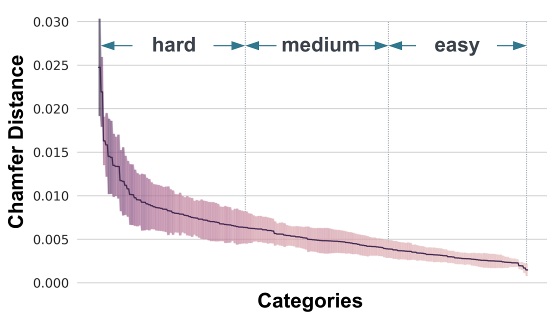

Figure 5. **Performance distribution of dense-view surface reconstruction.** Different categories are quite different. The colored area denotes a smoothed range of results.

MVSNeRF lags behind pixelNeRF on visual performance as we take 10 test frames widely distributed around the object in 360° by FPS sampling algorithm, where cost volume will be inaccurate on large-range viewpoint change. After further finetuning MVSNeRF for only 15 minutes on each test scene, MVSNeRF$_{\text{All*-ft}}$ achieve the best view synthesis results, comparable to test-time optimized NVS methods on nearby views. It is promising to utilize OmniObject3D, our large-scale and category-prosperous database, to build a standard benchmark suite for evaluation of diverse cross-scene NVS methods.

4.3. Neural Surface Reconstruction

A precise surface reconstruction from multi-view images enables a broad range of applications. For a single scene with dense-view images, algorithms are expected to conduct accurate, robust, and efficient surface reconstruction. When only sparse-view images are available, it is crucial to learn generalizable priors from a set of scenes or make use of other geometric cues to assist reconstruction. To this end, we study the two settings above for the surface reconstruction algorithms, accordingly.

Dense-View Surface Reconstruction. Under this setting, we include three representative methods: NeuS [55] and VolSDF [64] are two well-known approaches that bridge neural volume rendering with implicit surface representation. We also involve a voxel-based method called Voxurf [58], which leverages a hybrid representation for acceleration and fine geometry reconstruction.

Previous approaches in this task mainly perform evaluations and comparisons on 15 scenes from the DTU [1] dataset, which is not comprehensive and robust enough to demonstrate the ability of the methods in different scenarios. In comparison, we select three objects per category to run each of the three methods above for over 1,500 reconstructions in total, and we calculate the Chamfer Distance (CD) between the reconstructed surface and the ground truth. The distribution of the results is shown in Fig. 5. The average curve is obviously long-tailed: hard categories usually include low-textured and complex shapes(*e.g.*, vase, kennel, fork, bowl, cabinet, durian). We thus split the categories into three levels of "difficulty" based on the average curve, and the level-wise results are presented in Table 5. We can observe a clear margin among results in different levels for each method, indicating the split subsets to be generic and faithful.

Sparse-View Surface Reconstruction Dense-captured im-

Table 6. Sparse-view (3 views) surface reconstruction results.

Method	Train	Chamfer Distance $\times 10^3$ (↓) Hard	Medium	Easy	Avg
NeuS [55]	Single	29.35	27.62	24.79	27.3
MonoSDF [67]	Single	35.14	35.35	32.76	34.68
SparseNeuS [31]	1 cat.	34.05	31.32	31.14	32.36
	10 cats.	30.75	30.11	28.37	29.87
	All cats. Easy	28.39	**26.65**	23.76	26.48
	All cats. Medium	**27.38**	26.66	**23.08**	**25.87**
	All cats. Hard	27.42	26.95	24.63	26.47
MVSNeRF [7]	All cats.	56.68	48.09	48.70	51.16
pixelNeRF [66]	All cats.	63.31	59.91	61.47	61.56

ages of a scene are sometimes not available, so we study the sparse-view scenario here. The following methods are included: NeuS [55] with sparse-view input; MonoSDF [67], which takes in geometry cues estimated by pretrained models; SparseNeuS [31], a generic surface prediction pipeline that learns generalizable priors; pixelNeRF [66] and MVSNeRF [7] from Sec. 4.2, whose geometries are extracted from the density field. For NeuS, MonoSDF, and pixelNeRF, we use Farthest Point Sampling (FPS) to sample views that are most widely distributed; for SparseNeuS and MVSNeRF, we conduct FPS among the nearest 30 camera poses from a random reference view. A total of three views are sampled in each experiment.

The quantitative and qualitative comparisons are shown in Table 6 and Fig. 4, respectively. We first observe clear artifacts in all the sparse-view reconstructed results. Among them, SparseNeuS trained on enough data leads to the best quantitative performance on average, and the pre-division does not result in an obvious difference across difficulty levels. NeuS with sparse-view input achieves a surprisingly good performance. As shown in Fig. 4, the FPS sampling equips NeuS with a coherent global shape, especially for thin structures like case 3, while it also encounters severe local geometry ambiguity like case 1. MonoSDF, on the contrary, partially overcomes the issue of ambiguity via the assistance of predicted geometry cues in this case. However, it relies strictly on the accuracy of the estimated depth and normal, and thus easily fails when the estimation is inaccurate (*e.g.*, case 2 and 3). The surfaces extracted from generalized NeRF models, *i.e.*, pixelNeRF and MVSNeRF, are of relatively low quality.

In brief, the critical problem of sparse-view surface reconstruction has not been well solved currently. OmniObject3D is a promising database to study generalizable surface reconstruction pipelines as well as strategies for a robust usage of estimated geometry cues.

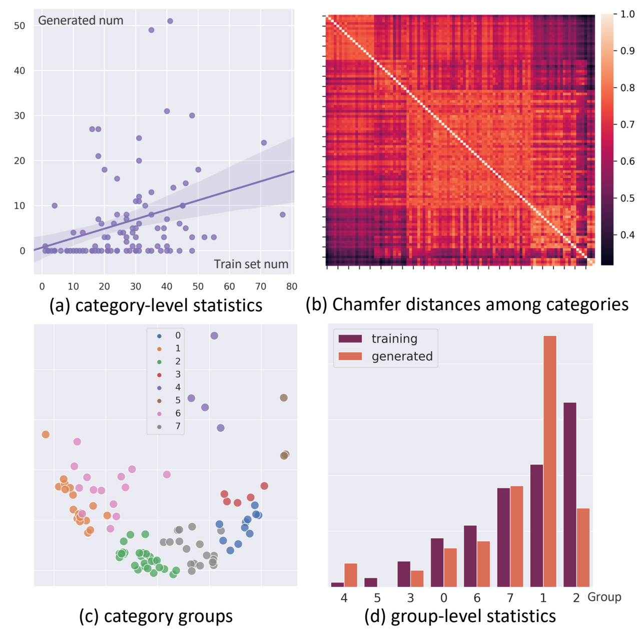

Figure 6. **The category distribution of the generated shapes. (a)** shows a weak positive correlation between the number of generated shapes and training shapes per category. **(b)** visualizes the correlation matrix among different categories by Chamfer Distance between their mean shapes. **(c)** visualizes categories being clustered into 8 groups by KMeans. **(d)** presents a clear training-generation relation in the group-level statistics.

4.4. 3D Object Generation

In this section, we adopt a state-of-the-art generative model that directly generates explicit textured 3D meshes, namely GET3D [17]. GET3D is originally evaluated on six categories (*Car, Chair, Motorbike, Animal, House, and Human Body*) with independent models, the number of shapes in each category ranging from 337 to 7,497. In comparison, OmniObject3D contains much more categories with a smaller number of objects in each, and thus is supposed to be jointly learnt with multiple categories in one model.

We first provide some qualitative results in Fig. 7, where we show various textured meshes generated by two models: one trained on 100 randomly selected categories and another trained on a subset of toys. The textures are rather realistic, and the shapes are coherent, enhanced by fine geometry details (*e.g.* lychee, pineapple). We explore the latent space of the model and show interpolation results in Fig. 8. We can observe a smooth transition across instances that are semantically different. We would like to further analyze the performance of the generative model trained on OmniObject3D from three aspects, *i.e.*,*semantic distribution*, *diversity*, and *quality*.

Semantic Distribution. We randomly select 100 categories to jointly train an unconditional model. At infer-

Figure 7. **Examples of the generated textured shapes rendered in Blender.** OmniObject3D enables GET3D with realistic generation across a wide range of categories.

Figure 8. **Shape interpolation.** We interpolate both geometry and texture latent codes from left to right.

Table 7. Quantitative evaluations on different data splits.

Split	#Objs	#Cats	Cov (% ↑)	MMD (↓)	FID (↓)	FID^{ref}
Furniture	265	17	**67.92**	4.27	87.39	58.40
Fruits	610	17	46.72	3.32	105.31	87.15
Toys	339	7	55.22	**2.78**	122.77	41.40
Rand-100	2951	100	61.70	3.89	**46.57**	**8.65**

ence time, we randomly generate 1,000 textured meshes and ask human experts to label them with the corresponding 100 classes. Shapes that cannot be easily classified are not counted. Fig. 6 (a) shows that the number of generated shapes per category is highly imbalanced, and it has a weak positive correlation with the corresponding training scale. However, the categories are not independent but rather highly correlated. We calculate the "mean shape" for each category and visualize the Chamfer Distance among them in Fig. 6 (b). The distance matrix indicates that the categories potentially to be further split into several groups. Regarding each row of the matrix as a feature vector, we use KMeans to cluster the categories into 8 groups (Fig. 6 (c)) and carry out a group-level statistics in Fig. 6 (d). It demonstrates a clear trend that the generated number of shapes increases along with or even faster than the training number of shapes in the group. This reveals that the train-set bias is enlarged during generation. However, it also depends on the inner-class or inner-group divergence. For example, Group-2 (883 instances in 27 categories) has the largest number of training samples, while it suffers from a high divergence among categories and fails to dominate the generated shapes; Group-1 (587 instances in 18 categories) has a relatively small divergence, which becomes the most popular in the generated shapes. More details can be found in the supplementary material.

Diversity and Quality. We select four representative data subsets for training, namely *fruits*, *furniture*, *toys*, and *Rand-100*. We randomly split each subset to training (80%) and testing (20%). We leverage three metrics for a quantitative evaluation. For geometry, we use Chamfer Distance (CD) to compute the Coverage score (Cov) and Minimum Matching Distance (MMD), which focus on the diversity and quality of the shapes, respectively; for texture, we adopt the widely-used FID [21]. But the FID metrics computed with different test splits are not directly comparable, and it bears a large variance when the test set is small. We thus introduce a reference for each subset, namely FID^{ref}, which is the FID between the train and test set. The results are shown in Table 7. *Furniture* suffers from the lowest quality (MMD) since the small train set with 17 categories is a difficult training source. *Fruits* has the same number of categories, while being 2.3 times larger in scale, and some fruits share a very similar structure. This leads to relatively higher quality and lower diversity (Cov). *Toys* achieve the best quality by training on only 7 categories. *Rand-100* is the most difficult case, and we can observe a trade-off between quality and diversity. The FID and FID^{ref} are high for the first three subsets because of the small testing set.

In brief, training generative models on a large-vocabulary and realistic dataset is a promising but challenging task. We reveal crucial problems like the semantic distribution bias and various exploration difficulties in different groups. Realistic and large-vocabulary 3D object generation still needs further examination.

5. Conclusion and Outlook

We introduce OmniObject3D, a large-vocabulary 3D object dataset with massive high-quality real-scanned 3D objects, including 6,000 objects from 190 categories. It provides rich annotations, including textured 3D meshes, sampled point clouds, posed multi-view images rendered by Blender, and real-captured video frames with foreground masks and COLMAP camera poses. We set up four evaluation tracks, revealing new observations, challenges, and opportunities for future research in realistic 3D vision.

Outlook. Beyond the four benchmarks, OmniObject3D can be further combined with 3D scenes to facilitate more holistic tasks in both 2D and 3D vision, automatically producing accurate annotations for a variety of user-defined tasks.

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